

Riccardo Orlando

PH.D. IN DEEP LEARNING AND NATURAL LANGUAGE PROCESSING

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Experience

Senior AI Engineer

PRINCIPLED INTELLIGENCE

Rome, Italy

Sept. 2026 - Current

AI Scientist

WORLD BANK GROUP

Rome, Italy

Mar. 2025 - Aug. 2026

- Owned the end-to-end design of an LLM-based event extraction system for monitoring food insecurity risk factors from over 150 million news articles, defining the synthetic data strategy, fine-tuning experiments, and evaluation benchmark for grounded structured extraction.
- Drove the research-to-production lifecycle of ImpactAI's core agentic pipeline, designing and deploying orchestration, evidence retrieval, and entity linking components to automate complex meta-analysis workflows.
- Owned the design of an RCT retrieval benchmark, coordinating with economists on data curation and labeling criteria to enable evaluation of retrieval and generation systems over scientific literature.

Researcher

SAPIENZA UNIVERSITY OF ROME

Rome, Italy

Nov. 2021 - Jan. 2025

- Led model pretraining and data preprocessing for Minerva (1B, 3B, and 7B) LLMs, the first family of Open Foundation Models trained from scratch on Italian data.
- Developed ReLiK, a novel retrieval-augmented system achieving SOTA performance in Entity Linking and Relation Extraction.
- Led the data annotation process that resulted in a novel dataset for Semantic Role Labeling.
- Developed production-ready pipelines for several state-of-the-art NLU tasks.
- Served as Teaching Assistant for MSc courses in Natural Language Processing.

NLP Engineer

BABELSCAPE

Rome, Italy

Feb. 2021 - Nov. 2021

- Designed and implemented end-to-end multilingual NLP pipelines for production deployment.
- Built scalable ML serving framework that drastically reduced model deployment time.
- Authored and presented two papers at EMNLP 2021.

Software Engineer

CAPGEMINI

Rome, Italy

Mar. 2018 - Sept. 2018

- Developed a back-end application using Spring Boot to handle internal workflows for different customers.
- Contributed to front-end development using Angular.

Education

Ph.D. in Computer Science (Deep Learning & Natural Language Processing)

SAPIENZA UNIVERSITY OF ROME

Rome, Italy

Nov. 2021 - Jan. 2025

- Thesis: Enhancing Semantic Understanding Across Multiple Dimensions: Towards A Unified Framework for Semantic Knowledge Extraction.
- Advisor: Roberto Navigli

Master of Science in Engineering in Computer Science

SAPIENZA UNIVERSITY OF ROME

Rome, Italy

Oct. 2018 - Jan. 2021

- Final Grade: 110/110 with honors
- Thesis: Automatic Approaches to Produce Multilingual Training Data for Semantic Role Labelling.

Bachelor of Science in Computer Science

SAPIENZA UNIVERSITY OF ROME

Rome, Italy

Oct. 2014 - May 2018

- Final Grade: 101/110
- Thesis: Densest Subgraph in Fork/Join.

Relevant Publications

2025

Karim Ghonim, Pere-Lluís Huguet Cabot, Riccardo Orlando, and Roberto Navigli. **RAED: Retrieval-Augmented Entity Description Generation for Emerging Entity Linking and Disambiguation.**

EMNLP 2025

2024	Francesco Maria Molfese, Simone Conia, Riccardo Orlando , and Roberto Navigli. ZEBRA: Zero-Shot Example-Based Retrieval Augmentation for Commonsense Question Answering.	EMNLP 2024
2024	Riccardo Orlando , Luca Moroni, Pere-Lluís Huguet Cabot, Edoardo Barba, Simone Conia, Sergio Orlandini, Giuseppe Fiameni, and Roberto Navigli. Minerva LLMs: The First Family of Large Language Models Trained from Scratch on Italian Data.	CLiC-it 2024
2024	Riccardo Orlando , Pere-Lluís Huguet Cabot, Edoardo Barba, and Roberto Navigli. ReLiK: Retrieve and Link, Fast and Accurate Entity Linking and Relation Extraction on an Academic Budget.	ACL 2024
2024	Simone Conia, Edoardo Barba, Abelardo Carlos Martínez Lorenzo, Pere-Lluís Huguet Cabot, Riccardo Orlando , Luigi Procopio, Roberto Navigli. MOSAICo: a Multilingual Open-text Semantically Annotated Interlinked Corpus.	NAACL 2024
2023	Riccardo Orlando , Simone Conia, and Roberto Navigli. Exploring Non-Verbal Predicates in Semantic Role Labeling: Challenges and Opportunities.	ACL 2023
2022	Riccardo Orlando , Simone Conia, Stefano Faralli, and Roberto Navigli. Universal Semantic Annotator: the First Unified API for WSD, SRL and Semantic Parsing.	LREC 2022
2021	Riccardo Orlando , Simone Conia, Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. AMuSE-WSD: An All-in-one Multilingual System for Easy Word Sense Disambiguation.	EMNLP 2021
2021	Simone Conia, Riccardo Orlando , Fabrizio Brignone, Francesco Cecconi, and Roberto Navigli. InVeRo-XL: Making Cross-Lingual Semantic Role Labeling Accessible with Intelligible Verbs and Roles.	EMNLP 2021

Skills

Frameworks	Deep Learning & Machine Learning (PyTorch, PyTorch Lightning, Hydra, Transformers, ONNX, TensorFlow, scikit-learn)
Back-end	FastAPI, Django, REST API
Languages	Python, Java, Kotlin, Bash/Shell
Developer Tools	Git, Docker, VS Code, Visual Studio, PyCharm, IntelliJ

Honors & Awards

2024	Winner , 3x Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type C for compute hours at CINECA, the Italian HPC Center	Rome, Italy
2023	Winner , Italian SuperComputing Resource Allocation (ISCRA) Call for Proposal type B for compute hours at CINECA, the Italian HPC Center	Rome, Italy
2022	Winner , Call for Proposals for Scientific Research	Rome, Italy

Program Committees

2026	Reviewer , ARR March 2026, ARR May 2026
2025	Reviewer , ARR May 2025
2024	Reviewer , ARR February 2024, ARR June 2024
2023	Reviewer , ARR December 2023, ARR October 2023, EMNLP 2023, ACL 2023, EACL 2023
2022	Reviewer , LREC 2022
2021	Reviewer , EMNLP 2021